

Python Programming Assignment - 24

(1) What is lambda function in python? How is it different from a normal function?

→ A lambda function is a small anonymous function in python. normal function defines a reusable, multi-line function with a name and lambda defines a temporary, single expression function.

(2) What are the limitations of lambda functions?

→ In lambda functions can not use assignments ($x=2$) and can not use loops. this is the two major drawbacks.

(3) Explain the working of the map() function with an example?

→ map() function applies a function to every element in an iterable and returns a new sequence.

example:

```
nums = [1, 2, 3, 4]
```

```
Data = list(map(lambda x: x+1, nums))
```

```
print(Data) # [2, 3, 4, 5]
```

(4) How does map() differ from using a for loop?

→ map() is functional transformation tool that builds a new sequence automatically and loop is control flow statement to execute a block of code step-by-step.

<5> what is the filter function and when should it be used?

→ filter() function is used to extract elements from a iterable that satisfy a given condition. It works by applying a function to each element and keeping only those for which function returns true.

<6> Difference between map() and filter()

→ The core difference between is that map() modifies every item in a sequence while filter() select a subset of items by removing the ones that do not meet the conditions.

<7> what is a reduce()? why is not a built-in function in python?

→ reduce() applies a function to reduce the iterable to a single value. reduce() was moved from functools so that beginners wouldn't be tempted to write overly complex code, forcing dev to look for simpler, cleaner.

<8> Explain how reduce() works internally?

→ reduce(function, iterable[, initializer])

function: Takes argument and return single value.

iterable: The sequence to be reduced (list, Tuple, etc..)

initializer: A starting value that is placed before first element

returns: returns a single final value.

used? (9) Can lambda functions be used with map, filter & reduce together? Explain!

→ Yes lambda create short, anonymous, single-expression functions inline without using def keyword.
filter() extracts elements from an iterable that satisfy a specific true/false condition.
map() transforms every individual item within an iterable.
reduce() take arguments and return final single value.

(10) When should functional programming be preferred over procedural style in Python.

→ The primary difference is that procedural programming focuses on a sequence of ~~set~~ step-by-step instructions. and functional programming focuses on evaluating pure mathematical function without changing state or modify data.

Procedural

```
numbers = [1, 2, 3, 4, 5]
result = []
for num in numbers:
    if num % 2 == 0:
        result.append(num * 2)
print(result)
```

functional

```
numbers = [1, 2, 3, 4, 5]
result = [num ** 2 for num in
          numbers if num % 2 == 0]
print(result)
```